



Science

Stage 8

Paper 1

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

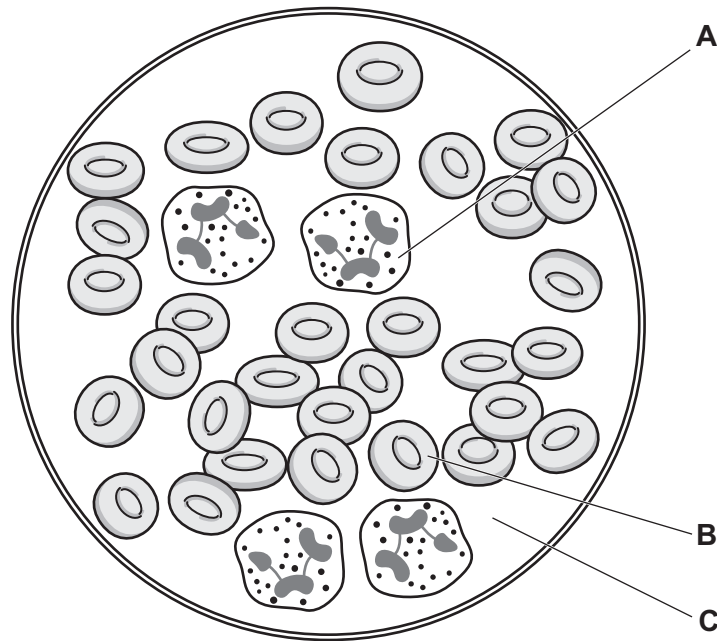
- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Pierre uses a microscope to look at some blood.

He draws an image of what he observes.



- (a) Complete the table about the functions of the components of the blood.

function	letter
transports nutrients	
transports oxygen	
protects against pathogens	

[2]

- (b) Pierre says,

'I predict that animals with a larger mass will have red blood cells with a larger diameter.'

He decides to look for secondary information from the internet.

- (i) Give **one advantage** and **one disadvantage** of using secondary information from the internet.

advantage

.....

disadvantage

.....

[2]

(ii) Here is the information Pierre finds.

animal	average mass of an adult in kg	average diameter of red blood cells in μm
dog	25	7.0
horse	300	5.7
sheep	80	4.5
goat	90	3.2
human	65	7.1
cat	4.5	5.8
rabbit	1.5	6.1

Tick (✓) to show if this information supports Pierre's prediction.

yes ☐ no ☐

Explain your answer.

Use information from the table.

.....

.....

.....

[1]

(c) Complete the sentence.

Choose from the list.

breathing

diffusion

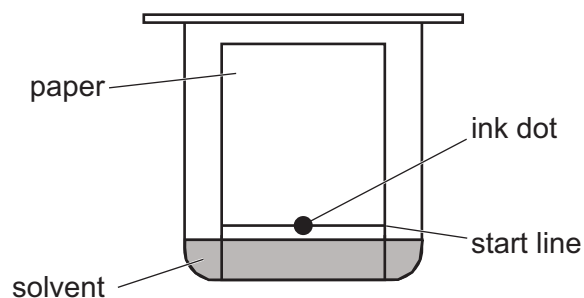
digestion

respiration

Gases move between blood and the air in the lungs by [1]

2 Carlos investigates the colours in ink.

(a) The diagram shows his equipment at the start of the investigation.



(i) Name this method of separating the colours in ink.

..... [1]

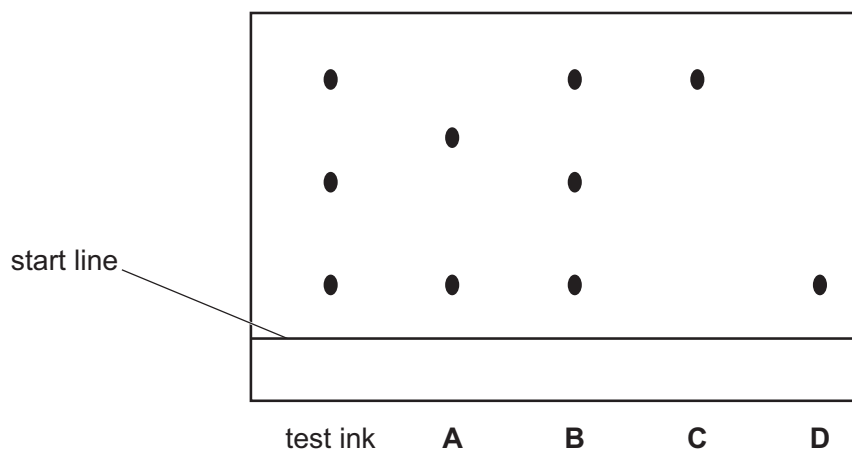
(ii) Explain why the ink dot must be above the solvent.

.....

 [2]

(b) Carlos separates the colours from five different samples of ink.

Look at the diagram that shows the results of these separations.



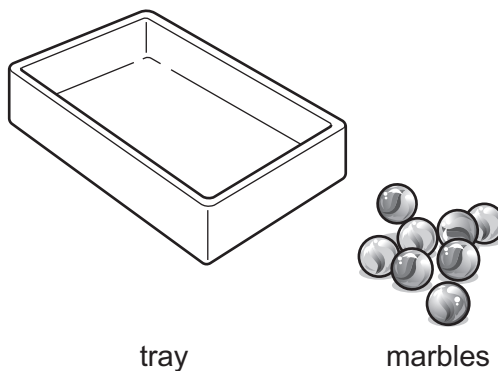
Which ink contains the same colours as the test ink?

Choose from **A**, **B**, **C** or **D**

Explain your answer.

.....
 [2]

- 3 Aiko uses a tray and some marbles to model gas pressure **in a container**.



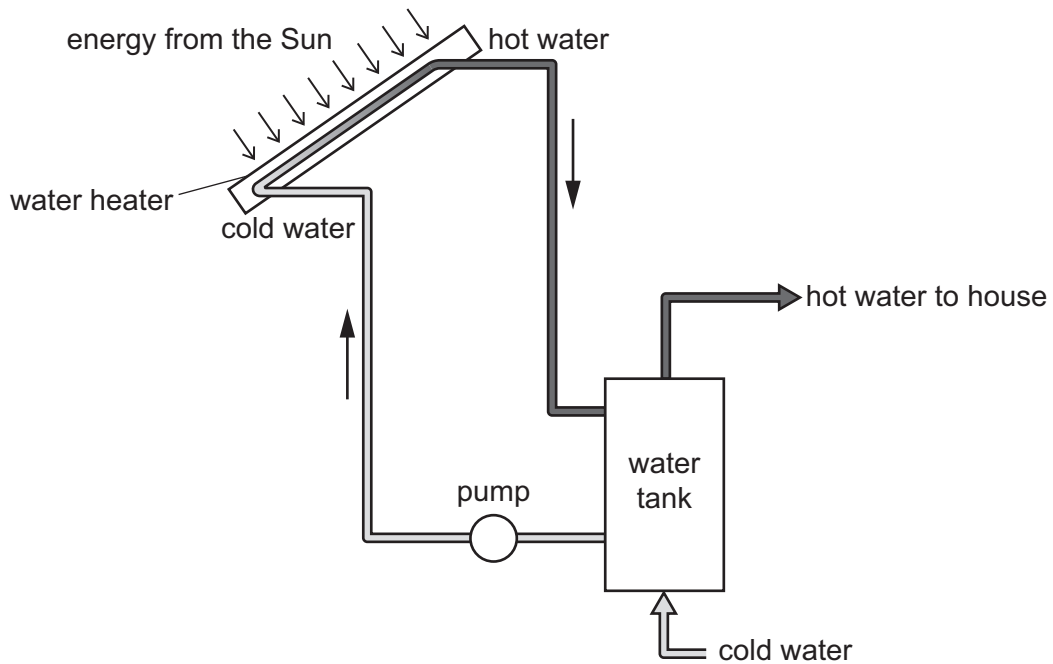
Explain how Aiko uses the equipment to model gas pressure **in a container**.

.....

.....

..... [2]

- 4 The diagram shows a water heating system that uses energy from the Sun.



- (a) (i) Write down **one** reason why this is an example of using a renewable energy resource.

.....
 [1]

- (ii) Suggest **one disadvantage** of using this method to heat water.

.....
 [1]

- (iii) Write down **one other renewable** energy resource.

..... [1]

- (iv) Write down **one non-renewable** energy resource.

..... [1]

(b) The picture shows a container made from plant material.



Which word describes this type of renewable material?

Circle the correct answer.

biochemical

bioplastic

cytoplasm

mineral

[1]

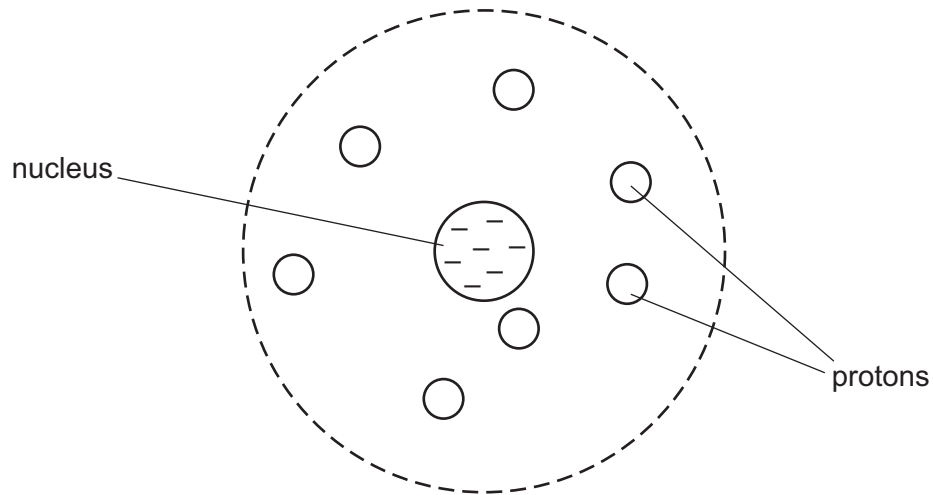
- 5 A balanced diet contains many different nutrients.

Draw a straight line from each **nutrient** to its correct **function in the body**.

nutrient	function in the body
calcium	for healthy eyesight
carbohydrate	to prevent dehydration
vitamin A	to make strong bones and teeth
protein	to provide energy and to be used as an energy store
water	for tissue growth and repair

[3]

6 Lily draws a diagram to show the model of an atom.



(a) (i) The diagram is **not** correct.

Write down **two** ways the diagram is **not** correct.

1

2

[2]

(ii) This model is named after a scientist.

Write down the name of this scientist.

..... [1]

(b) Explain how the particles in an atom are held together.

.....

.....

..... [2]

7 This question is about asteroids.

(a) The table shows some information about asteroids that have passed near to the Earth.

asteroid	date asteroid was nearest to the Earth	nearest distance from the Earth in km
A	February 2004	5481
B	March 2004	6542
C	October 2009	6259
D	October 2019	6242
E	August 2020	2946

(i) Which asteroid passed **nearer** than 5000 km from the Earth?

.....

[1]

(ii) Suggest why the distances in the table may **not** be accurate.

.....

.....

[1]

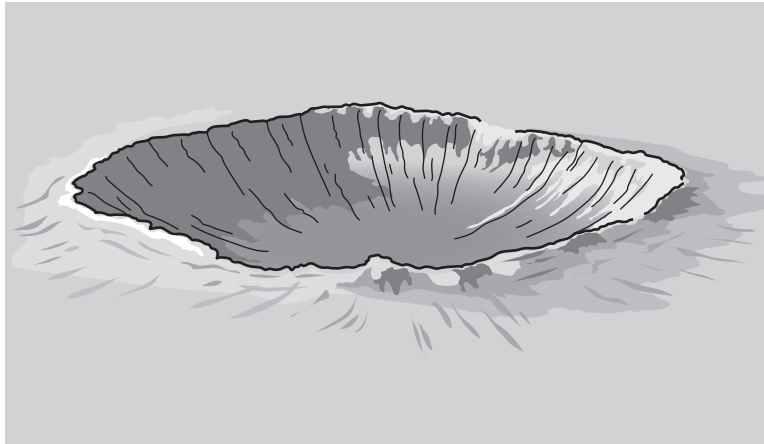
(iii) Describe how asteroids in space are formed.

.....

.....

[1]

- (b) The Moon has many craters formed when asteroids hit the surface of the Moon.

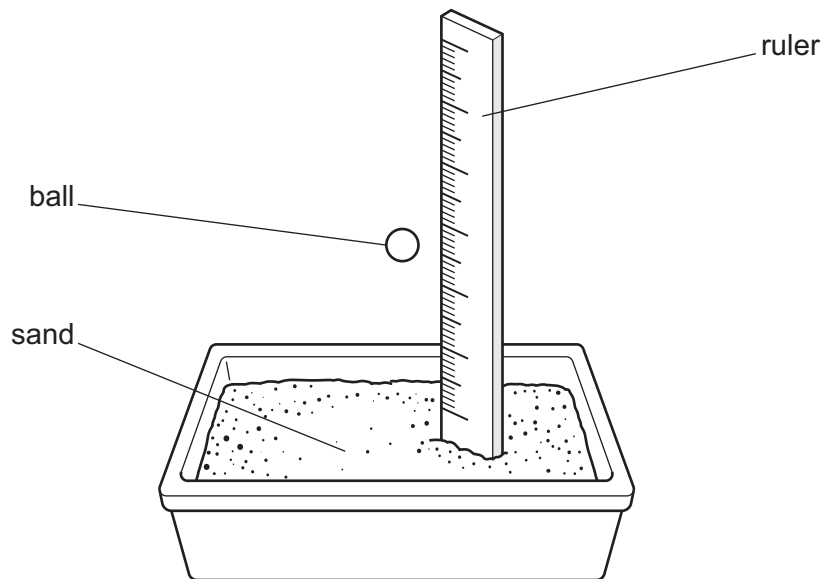


Ahmed makes this hypothesis about craters.

**‘The bigger the asteroid hitting the Moon,
the bigger the crater that is made.’**

Ahmed wants to test this hypothesis.

The diagram shows some of the equipment he uses.



Ahmed drops the ball and measures the size of the crater.

Describe how Ahmed uses this equipment to test his hypothesis.

.....

.....

..... [2]

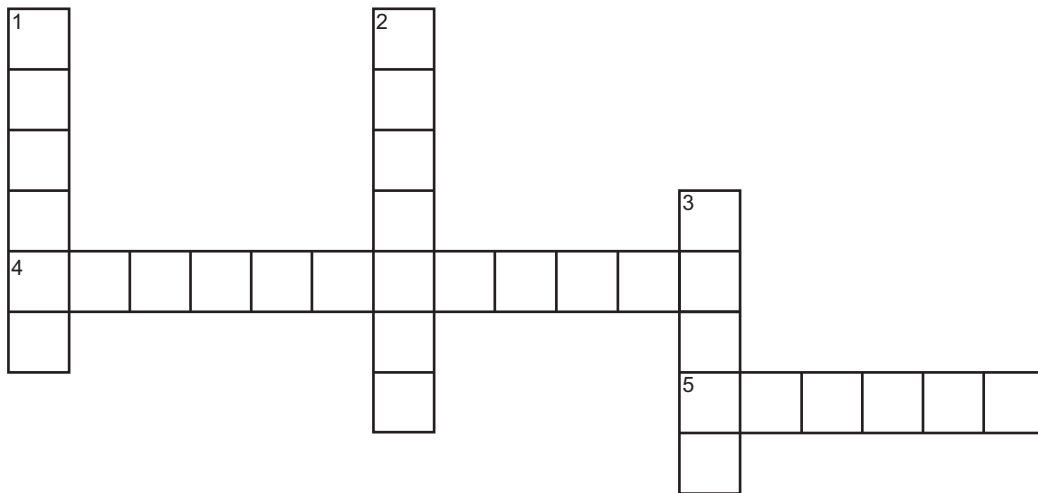
- 8 Safia makes an electromagnet.

Describe how Safia makes an electromagnet.

You may want to draw a diagram to help you answer the question.

[2]

9 Complete the crossword about aerobic respiration.



Clues

1 down

Which component of blood transports carbon dioxide?

.....

2 down

Which compound is a reactant of aerobic respiration?

.....

3 down

Which compound is a waste product of aerobic respiration?

.....

4 across

What is the name of the structure in a cell where aerobic respiration takes place?

.....

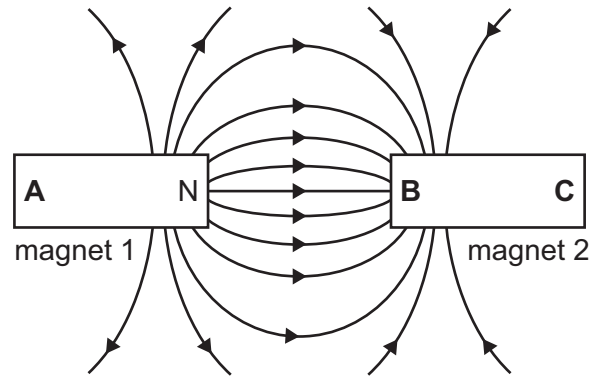
5 across

What is released during aerobic respiration?

.....

[4]

10 Blessy places two magnets next to each other as shown in the diagram.



The north pole, N, is labelled on magnet 1.

The other poles are labelled **A**, **B** and **C**.

(a) (i) What is shown by the pattern of lines around the two magnets?

..... [1]

(ii) Blessy writes some statements about the two magnets.

Tick (✓) the **correct** statements.

statement	
A is a N pole	
B is a S pole	
C is a N pole	

[1]

(b) Blessy brings pole **D** and pole **E** of two other magnets close to each other.

The magnets push each other away.

Which sentences are possible explanations?

Tick (✓) **two** boxes.

Pole **D** and pole **E** are both S poles.

☐

Pole **D** is a N pole and pole **E** is a S pole.

☐

Pole **D** is a S pole and pole **E** is a N pole.

☐

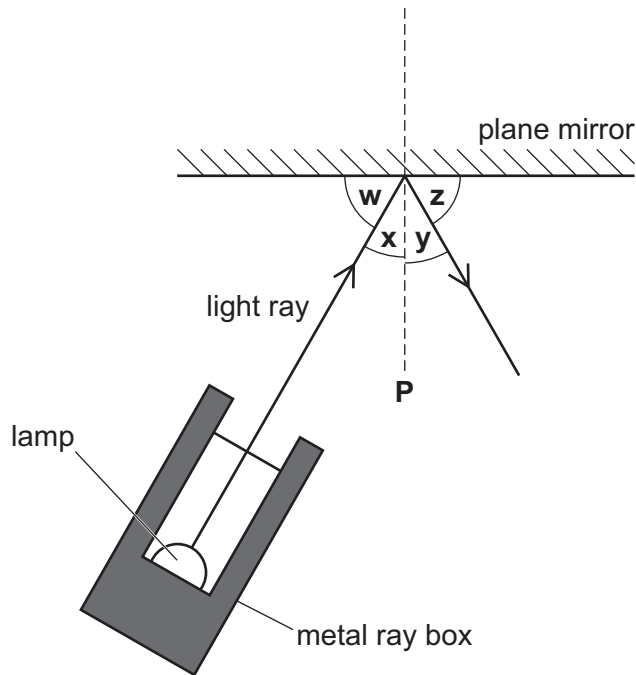
Pole **D** and pole **E** are both N poles.

☐

[2]

11 Mike investigates the reflection of a light ray at a plane mirror.

Look at his ray diagram.



(a) (i) Write down the name of the dotted line **P** in the diagram.

..... [1]

(ii) Which row in the table identifies the angle of incidence and the angle of reflection in Mike's diagram?

	angle of incidence	angle of reflection
A	w	x
B	w	y
C	x	y
D	x	z

row

[1]

- (b) Mike changes the angle of incidence and measures the angle of reflection.

The table shows the angles of incidence he uses.

Complete the table to predict the angles of reflection.

angle of incidence in °	angle of reflection in °
20	
30	
40	
50	

[1]

- (c) Mike uses a metal ray box connected to a power supply in his investigation.

Write down **one** safety hazard in Mike's investigation.

Describe how he avoids this hazard.

safety hazard

description of how Mike avoids the hazard

.....

[2]

12 Mia investigates solubility.

In her first experiment Mia:

- step 1 - pours 50 cm³ of water into a beaker
- step 2 - records the temperature of the water
- step 3 - adds 1g of a salt to the water
- step 4 - stirs the water until the salt dissolves
- step 5 - repeats steps 3 and 4 until no more salt dissolves
- step 6 - records the total mass of salt that dissolves.

Mia repeats the experiment four more times.

Each time she uses a different temperature of water.

Look at her results.

temperature of water in °C	mass of salt that dissolves in g
20	12
30	22
40	24
50	42
60	50

(a) What is the independent variable in this investigation?

..... [1]

(b) Describe the relationship between the temperature of the water and the mass of salt that dissolves.

.....
..... [1]

(c) Mia says,

‘I do not think my results are reliable.’

Write down **two** reasons why Mia is correct.

- 1
.....
- 2
.....

[2]

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The Periodic Table of Elements

Group																									
1	2	Key														3	4	5	6	7	8				
		atomic number atomic symbol name relative atomic mass																1 H hydrogen 1							
3 Li lithium 7	4 Be beryllium 9																	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
11 Na sodium 23	12 Mg magnesium 24																	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84								
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131								
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —								
87 Fr francium	88 Ra radium	89–103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganeson								



Science

Stage 8

Paper 1

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	C B A	2	all three correct = 2 marks one or two correct = 1 mark
1(b)(i)	(advantages) any one from easy to obtain cheap fast/quick way to get results no need to have the equipment to carry out experiments yourself (disadvantages) any one from quality/accuracy of data unknown could be biased unknown experimental method	2	each correct answer = 1 mark Accept any other reasonable response Accept any other reasonable response
1(b)(ii)	no ticked (no marks) and correct use of data from the table, e.g. the dog has less mass than the horse but has red blood cells with a larger diameter	1	if both boxes are ticked = 0 marks if neither box ticked accept no within the explanation Accept any correct example from the table

Question	Answer	Marks	Further Information
1(c)	diffusion	1	Accept any indication of the correct answer, e.g. ticking, circling or underlining, but answer line takes precedence

Question	Answer	Marks	Further Information
2(a)(i)	(paper) chromatography	1	
2(a)(ii)	any two from ink would dissolve in the solvent (idea that) solvent becomes stained with ink (idea that) ink spreads out not getting a clear result	2	each correct answer = 1 mark
2(b)	B (idea of) identical pattern of spots	2	each correct answer = 1 mark Note both have 3 spots is not sufficient

Question	Answer	Marks	Further Information
3	(Oliver) shakes marbles in the tray marbles collide with the side of the tray	2	each correct answer = 1 mark Ignore just marbles collide (with each other)

Question	Answer	Marks	Further Information
4(a)(i)	will not run out or is not finite	1	Ignore (idea that) can be used again
4(a)(ii)	(idea that) Sun may be covered by clouds or affected by weather or does not work in the dark or affected by daylight length or affected by seasons	1	
4(a)(iii)	wind or tidal	1	Accept any suitable example, e.g. wave or hydroelectric or geothermal or (renewable) biomass Do not accept nuclear
4(a)(iv)	natural gas or oil or fossil fuels or coal	1	Accept nuclear
4(b)	bioplastic	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information
5	nutrient calcium carbohydrate vitamin A protein water function in the body for healthy eyesight to prevent dehydration to make strong bones and teeth to provide energy and to be used as an energy store for tissue growth and repair	3	all five correct = 3 marks three or four correct = 2 marks one or two correct = 1 mark if more than one line from any nutrient to different functions = 0 marks for that nutrient

Question	Answer	Marks	Further Information
6(a)(i)	any two from nucleus should have a positive charge or nucleus should not have a negative charge proton label should be electron circles outside nucleus should have a negative charge	2	each correct answer = 1 mark Accept electrons are not arranged (in shells) Accept protons should be in the nucleus
6(a)(ii)	Rutherford	1	
6(b)	attraction between protons and electrons (idea of) attraction between positive and negative particles or electrostatic attraction	2	each correct answer = 1 mark Accept attraction between nucleus and electrons Note if no other marks awarded Accept attraction between opposite charges = 1 mark

Question	Answer	Marks	Further Information
7(a)(i)	E	1	
7(a)(ii)	(idea that) difficult to measure distance in space	1	
7(a)(iii)	formed from rocks left over from formation of a planetary system	1	
7(b)	repeats with different sized balls references to controlling a variable, e.g. drops ball from same height each time	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
8	connect wire to cell or connect wire to battery or connect wire to power supply wrap wire around a piece of iron (core or rod)	2	each correct answer = 1 mark Note marks may be awarded from a labelled diagram

Question	Answer	Marks	Further Information
9	¹P L A S ⁴M A I T O C H O N D R I ²G L U C S E ³W T ⁵E R N E R G Y	4	all five correct = 4 marks four correct = 3 marks three correct = 2 marks one or two correct = 1 mark Accept answers written on answer lines, but answers in the crossword take precedence

Question	Answer	Marks	Further Information
10(a)(i)	magnetic field	1	
10(a)(ii)	B is a S pole C is a N pole	1	both correct for the mark Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
10(b)	Pole D and pole E are both S poles. Pole D and pole E are both N poles.	2	each correct answer = 1 mark Note if one incorrect answer ticked = max 1 mark if two or more incorrect answers ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information										
11(a)(i)	normal (line)	1											
11(a)(ii)	C	1	Accept any indication of the correct answer, e.g. ticking, circling, underlining or x and y, but letter on answer line takes precedence										
11(b)	<table><tr><th>angle of incidence in °</th><th>angle of reflection in °</th></tr><tr><td>(20)</td><td>20</td></tr><tr><td>(30)</td><td>30</td></tr><tr><td>(40)</td><td>40</td></tr><tr><td>(50)</td><td>50</td></tr></table>	angle of incidence in °	angle of reflection in °	(20)	20	(30)	30	(40)	40	(50)	50	1	all four correct = 1 mark
angle of incidence in °	angle of reflection in °												
(20)	20												
(30)	30												
(40)	40												
(50)	50												
11(c)	(safety hazard) ray box may be hot (description) leave to cool or use gloves/cloth or turn off when not in use	2	correct hazard = 1 mark correct description associated with the hazard = 1 mark Accept the equipment may be hot Accept (hazard) electric shock (description) keep away from water or turn power supply off when connecting ray box Note award marks for correct answers wherever written										

Question	Answer	Marks	Further Information
12(a)	temperature (of water)	1	
12(b)	higher the temperature the more salt dissolves	1	Accept ora
12(c)	results for 30 °C and 40 °C are too close together or results for 40 °C seems too low or results for 40 °C is an anomaly only one set of results or needs to repeat the investigation	2	each correct answer = 1 mark



Science

Stage 8

Paper 2

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

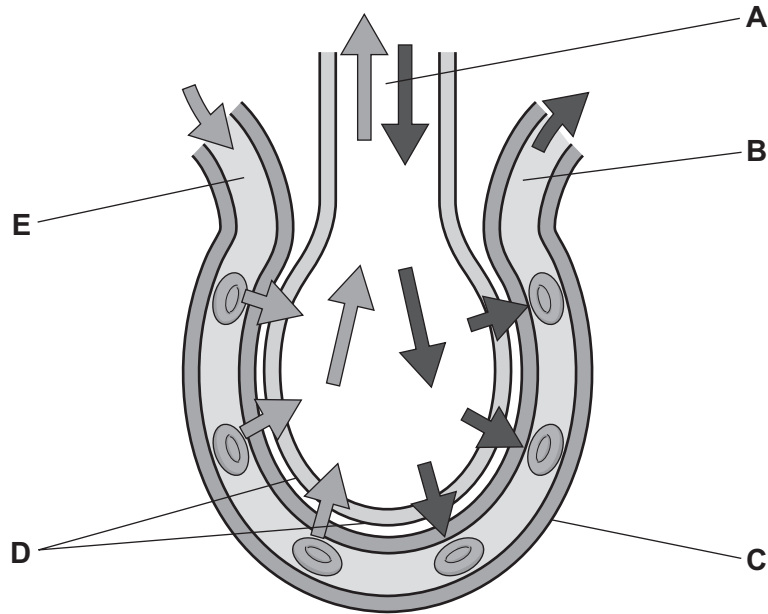
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INFORMATION

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1 Alveoli are air sacs found at the end of the bronchioles in the lungs.

(a) The diagram shows how gas is exchanged in alveoli.



Match the **description** to the correct **letter** shown in the diagram.

Write your answers in the table.

description	letter
place where the blood has the highest oxygen concentration	
blood capillary wall	
a surface where gas exchange happens	

[3]

(b) Alveoli are adapted to maximise gas exchange.

Describe **one adaptation** of the alveoli and explain how this helps to maximise gas exchange.

adaptation

.....

explanation

.....

[2]

- (c) Red blood cells transport oxygen around the body.

Which mineral in the diet is needed to make red blood cells?

..... [1]

2 This question is about reactions of magnesium.

(a) Mia heats a piece of magnesium over a blue Bunsen flame.



(i) The magnesium reacts with a gas in the air.

Write down the name of this gas.

..... [1]

(ii) Describe one **safety risk** in this investigation and how to control the risk.

safety risk

how to control risk

..... [2]

(b) Magnesium reacts with dilute hydrochloric acid.

Circle **all** the products of this reaction.

carbon dioxide

chlorine

hydrogen

magnesium chloride

magnesium hydroxide

water

[2]

- 3 Safia uses an electromagnet to pick up steel pins.

She changes the current in the wire and records her results in the table.

current in A	number of pins
0.25	1
0.50	2
0.75	3
1.00	1
1.25	5

- (a) Circle the anomalous result in the table.

[1]

- (b) The current in the wire changes the strength of the electromagnet.

Suggest **one other** factor that changes the strength of the electromagnet.

..... [1]

- 4 Space has many parts.

Draw a straight line from each **part** to its correct **description**.

part	description
galaxy	clouds of particles in space
	object orbiting a planet
solar system	large group of stars
	small planets
stellar dust	planets orbiting around a star

[3]

5 Some industries produce toxic substances.

These toxic substances sometimes enter food chains.

Look at the food chain.

algae → mayfly → small fish → big fish → otter

(a) Suggest a habitat where this food chain is found.

..... [1]

(b) The table shows how the amount of toxic substance increases along the food chain.

organism	relative concentration of toxic substance
algae	1
mayfly	12
small fish	50
big fish	320
otter	5000

(i) Explain why the relative concentration of toxic substance is **much higher** in an otter than a big fish.

.....
 [1]

(ii) The amount of toxic substance increases along the food chain.

What is this process called?

..... [1]

(c) Write down **one impact** of increasing the concentration of toxic substances on an ecosystem.

.....
 [1]

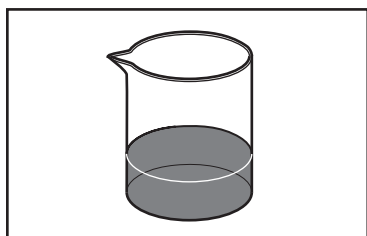
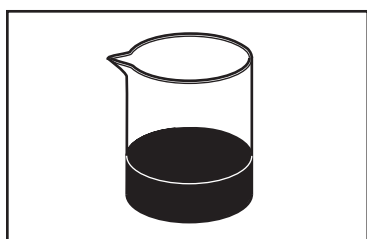
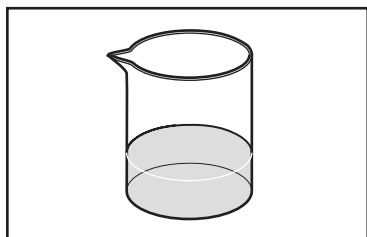
6 This question is about dissolving.

(a) When a blue food colour dissolves in water the solution turns blue.

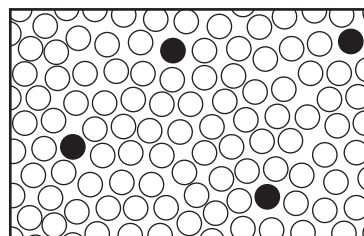
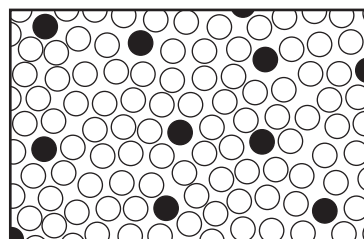
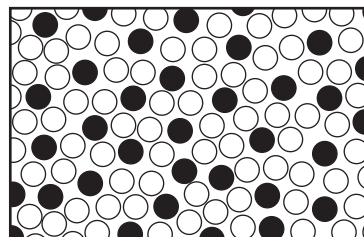
As the concentration of the blue food colour increases the solution becomes darker.

Draw a straight line from each **solution** to the correct **particle diagram** representing the solution.

solution



particle diagram



● = particle of blue food colour
○ = particle of water

[1]

(b) Circle the word in **bold** that makes each sentence correct.

Blue food colour dissolves in water.

The blue food colour is the **solute** / **solvent** / **soluble**.

Water is the **solute** / **solvent** / **soluble**.

[1]

7 Ahmed investigates primary and secondary colours.

(a) Which colour of light is produced when red light and green light are mixed?

Circle the correct answer.

yellow

blue

cyan

magenta

white

[1]

(b) Ahmed wears a magenta and green striped shirt.

Complete the sentences to explain how Ahmed's shirt looks in **blue light**.

In blue light, the magenta stripes look the colour

This is because magenta

.....

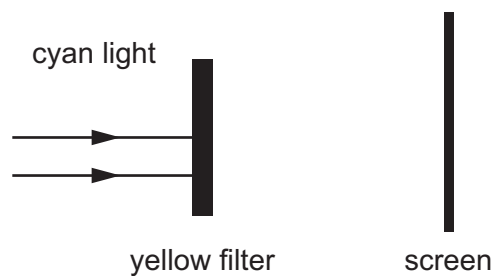
In blue light, the green stripes look the colour

This is because green

.....

[2]

(c) Ahmed passes cyan light through a yellow filter.



Explain which colour light is seen on the screen.

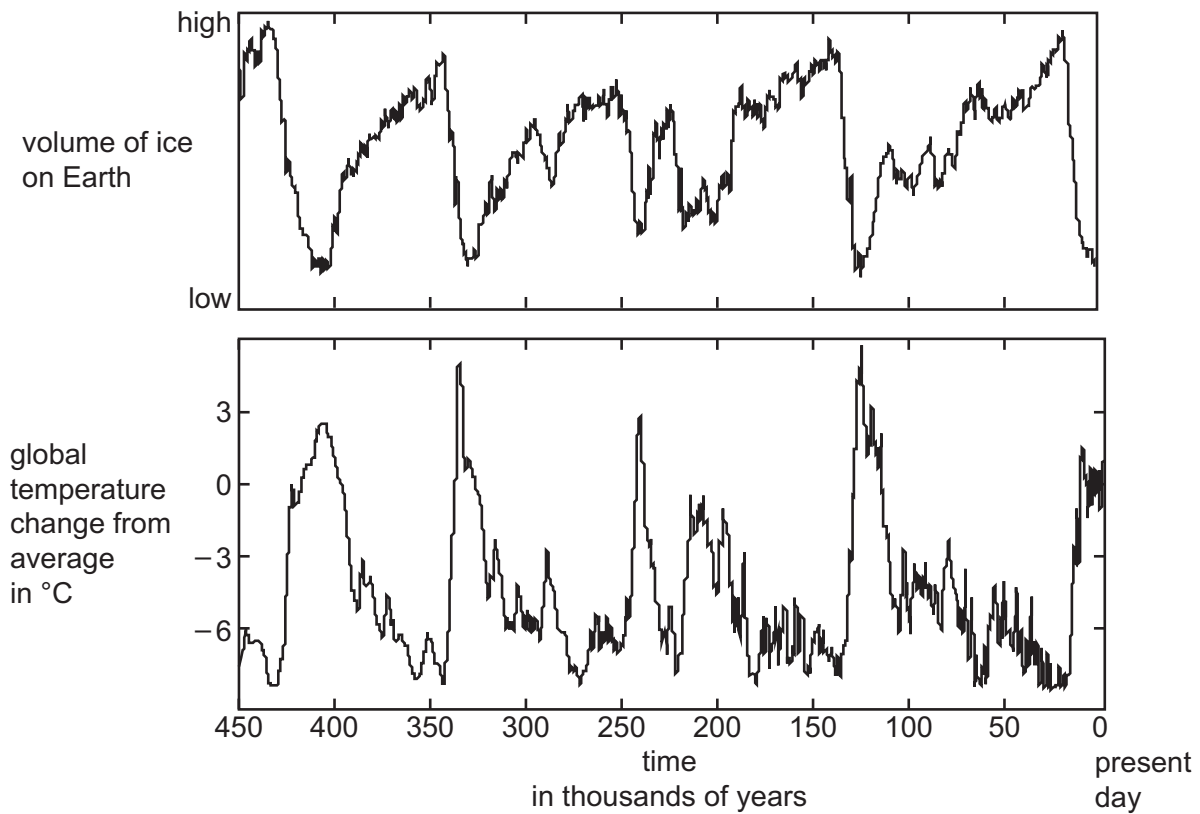
colour light

reason

.....

[1]

- 8 Look at the graphs about the volume of ice on Earth and global temperature change.



- (a) Use information from the graphs to answer these questions.

- (i) When was the **highest** global temperature?

..... thousand years ago [1]

- (ii) Describe the relationship between the global temperature and the volume of ice on Earth.

.....
 [1]

(b) Climate and weather are two different things.

Which statements are about **climate**?

Tick (✓) **all** the correct boxes.

average rainfall for a particular region	
changes in the atmosphere during a 24-hour period	
highest temperature recorded in a week	
long-term temperature pattern in a given area	
predicting when the next thunderstorm will happen	

[2]

9 Gold metal is used to make jewellery.

The purity of gold mixed with other metals is measured using a system called the carat rating.

The table shows the percentage of gold and the percentage of other metals in different carat ratings.

carat rating	minimum percentage of gold in the metal	maximum percentage of other metals mixed with the gold
24	100.0	0.0
22	91.6	8.4
18	75.0	25.0
14	58.5	41.5
10	41.7	58.3

A piece of gold mixed with other metals has a mass of 200 g.

The mass of gold in this mixture is 86.5 g.

(a) Calculate the percentage of gold in the mixture.

percentage of gold % [1]

(b) What is the carat rating for this mixture of gold and other metals?

carat rating [1]

10 Complete the sentences about diffusion.

Diffusion occurs in gases and

When perfume is sprayed in a room, the perfume particles and air particles move and

.....

Diffusion occurs more quickly if a gas is hotter because the gas particles move

.....

[2]

11 Yuri investigates the water content of potatoes.

Yuri:

step 1 - cuts a piece of potato with a sharp knife into a cube of mass 10.0 g

step 2 - heats the cube of potato in a microwave for 1 minute

step 3 - uses a balance to find the mass of the cube of potato

step 4 - repeats steps 2 and 3 until the cube of potato has been heated for a total of 10 minutes.

(a) Here are his results.

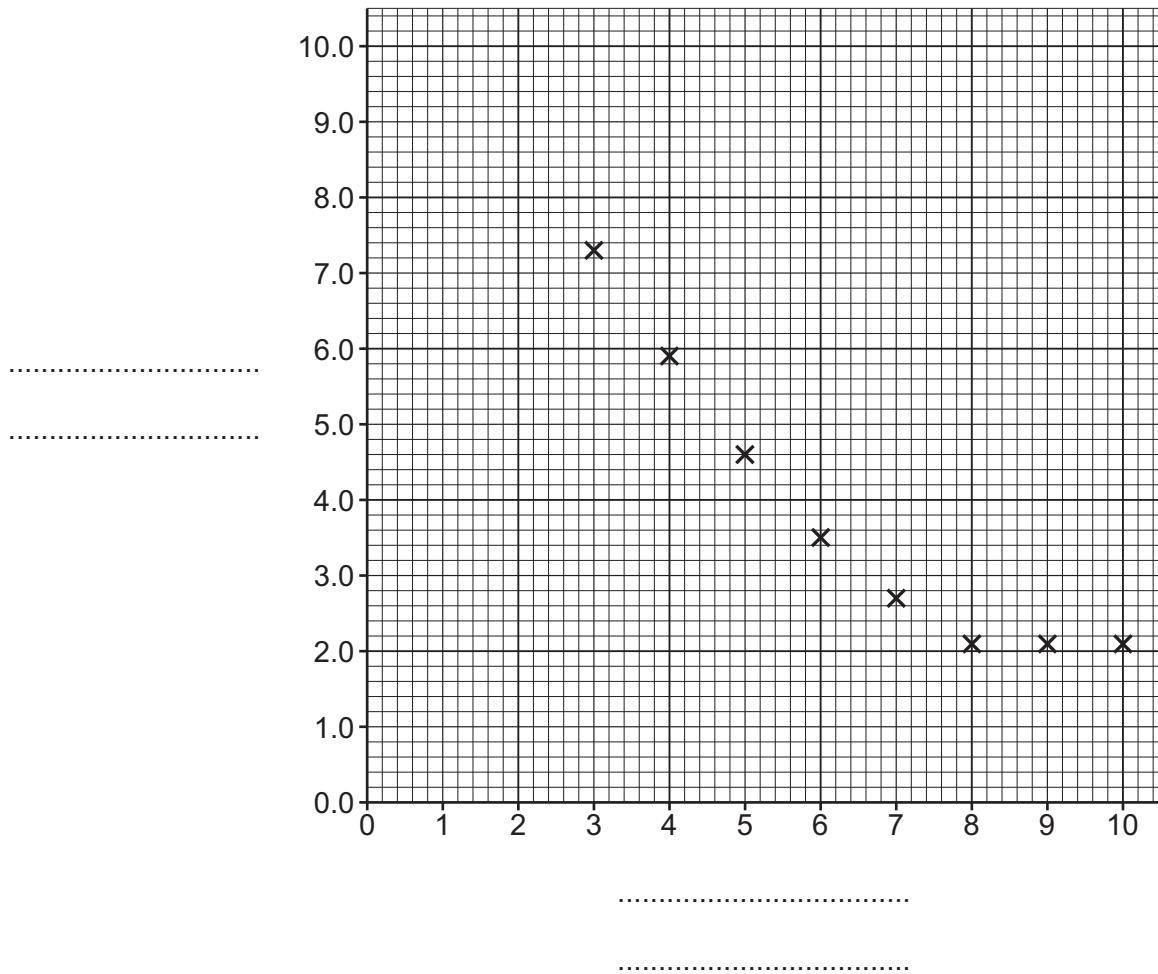
time in minutes	mass of cube of potato in g
0	10.0
1	9.4
2	8.5
3	7.3
4	5.9
5	4.6
6	3.5
7	2.7
8	2.1
9	2.1
10	2.1

Look at the graph of his results.

The last eight masses from the table have been plotted on the graph.

Complete the graph by:

- labelling the axes
- plotting the first three masses from the table on the graph
- drawing a curve of best fit.



[4]

(b) Yuri says,

**'I have read that vegetables are 90% water.
I predict 90% of this potato will be water.'**

Tick (✓) to show if the results support Yuri's prediction.

yes

☐

no

☐

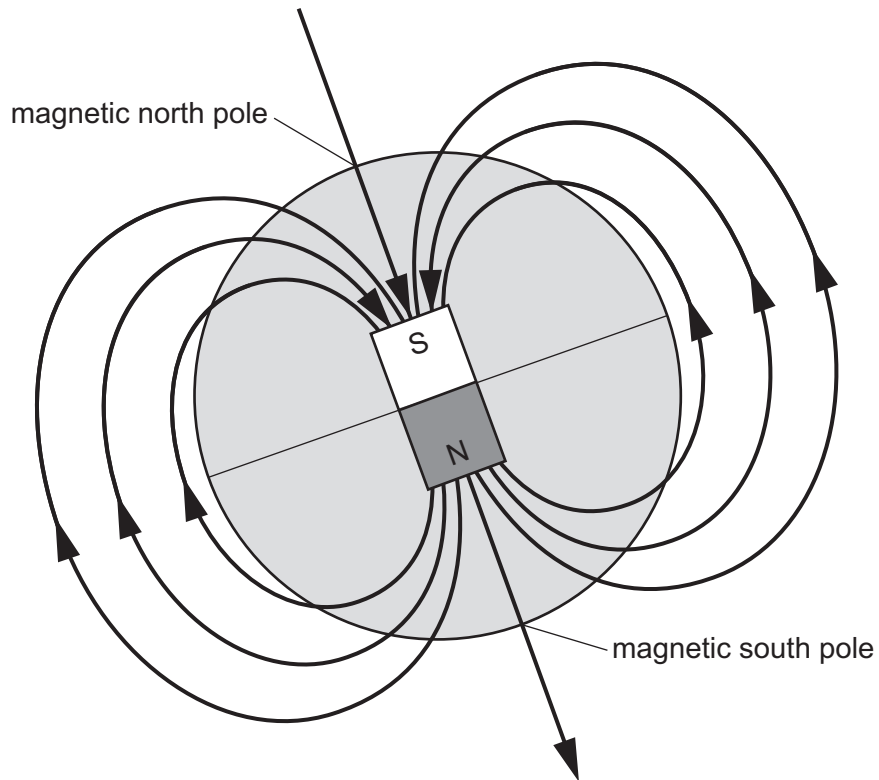
Explain your answer.

.....

.....

[1]

12 The diagram shows an analogy to describe the Earth's magnetism.



(a) Which part of the Earth's structure is represented by the bar magnet?

..... [1]

(b) What do the arrows represent?

..... [1]

13 Mike investigates reactions.

In his first experiment Mike:

- pours some water into a beaker
- measures the temperature of the water
- adds 1g of a solid to the water
- stirs the mixture until the temperature stops changing.

Mike repeats the experiment with four other solids.

Look at his results.

solid	temperature at the start in °C	temperature at the end in °C
A	20	20
B	21	28
C	19	15
D	18	19
E	21	27

(a) (i) How many solids have an endothermic reaction with water?

..... [1]

(ii) Mike has **not** done a fair test.

Suggest **one** way Mike could improve his investigation to make it a fair test.

.....
 [1]

(iii) Mike describes solid **A** as being inert.

What is meant by the term inert?

..... [1]

(b) One example of an exothermic reaction is adding calcium oxide to water.

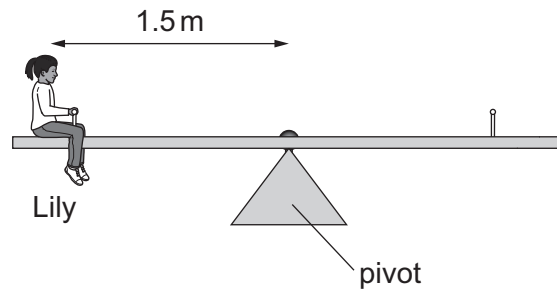
The product is calcium hydroxide.

Write the word equation for this reaction.

..... [1]

14 Lily and Blessy play on a seesaw.

(a) Lily sits at a distance of 1.5 m from the pivot.



Lily weighs 500 N.

Calculate the moment of Lily's weight about the pivot.

Include the unit in your answer.

moment of Lily's weight = unit [3]

(b) Blessy sits on the seesaw with Lily.

Blessy weighs 600 N.

The seesaw is balanced.

Calculate the distance Blessy sits from the pivot.

Use ideas about the principle of moments.

distance from pivot m [2]

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The Periodic Table of Elements

Group									
1	2	3					6	7	8
3 Li lithium 7	4 Be beryllium 9	Key atomic number name atomic symbol relative atomic mass					1 H hydrogen 1		
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	19 K potassium 39	20 Ca calcium 40
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —
		111 Ro roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
		119 Te tellurium 127	120 Po polonium —	121 At astatine —	122 Bi bismuth 209	123 Po polonium —	124 At astatine —	125 Rn radon —	126 Fr francium —
		127 I iodine 127	128 Xe xenon 131	129 Po polonium —	130 At astatine —	131 Rn radon —	132 Fr francium —	133 Og oganesson —	134 Ts tennessine —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

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Science

Stage 8

Paper 2

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

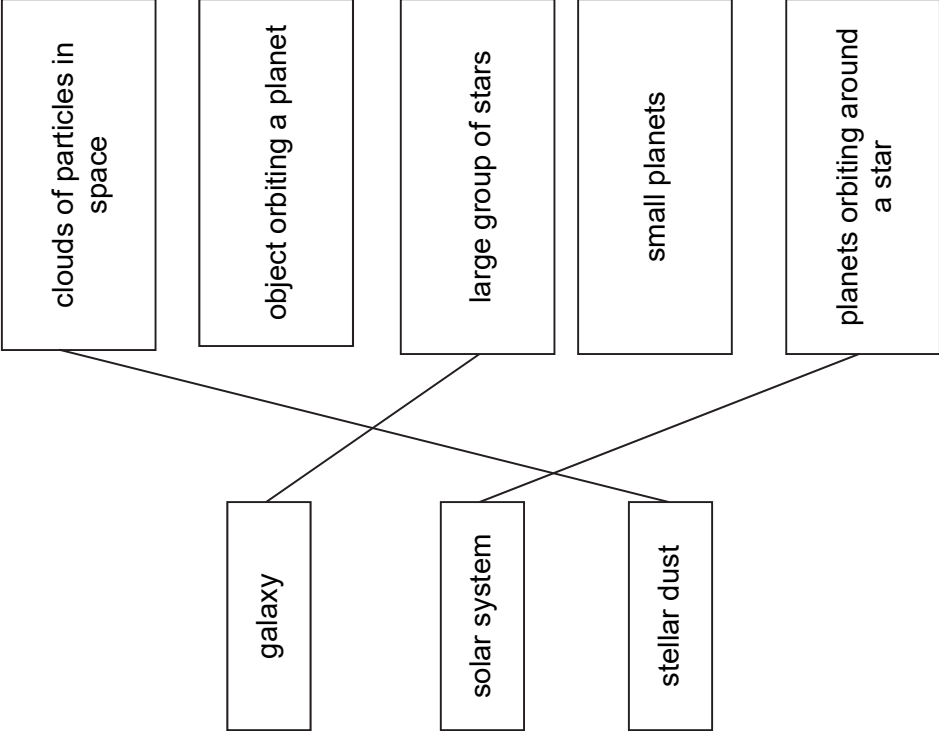
Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

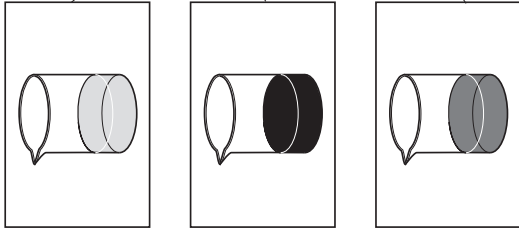
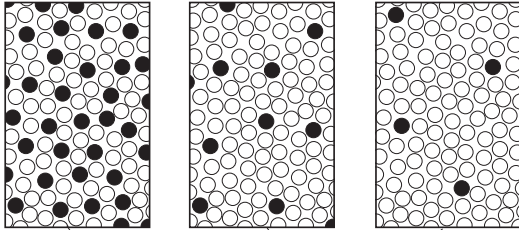
Question	Answer	Marks	Further Information
1(a)	B C D	3	each correct letter = 1 mark more than one letter in a box = 0 marks for that description
1(b)	(adaptation) thin wall (explanation) short diffusion distance or increases rate of diffusion or (adaptation) rich blood supply (explanation) to transport gases quickly or maintain diffusion gradient or increases rate of diffusion or (adaptation) large surface area (explanation) increases rate of diffusion or (adaptation) moist walls (explanation) so gases dissolve or increases rate of diffusion	2	any correct adaptation = 1 mark an explanation that is correctly linked to the adaptation = 1 mark Accept more efficient diffusion for increases rate of diffusion Accept more efficient diffusion for increases rate of diffusion Accept more efficient diffusion for increases rate of diffusion Accept more efficient diffusion for increases rate of diffusion Note award marks wherever written
1(c)	iron	1	Accept Fe

Question	Answer	Marks	Further Information
2(a)(i)	oxygen	1	Ignore O or O ₂
2(a)(ii)	(safety risk) burning metal (how to control risk) need eye protection or (safety risk) bright flame (how to control risk) don't look directly at flame or look through a blue filter or blue glass or (safety risk) (idea that) equipment will be hot (how to control risk) wear heat protective gloves	2	a correct safety risk = 1 mark control correctly linked to the risk = 1 mark Accept named eye protection, e.g. goggles Ignore just gloves Ignore other safety controls, e.g. do not run or do not eat
2(b)	hydrogen magnesium chloride	2	each correct answer = 1 mark Accept one circle around both words. three circles two correct = 1 mark three circles one correct = 0 marks more than three answers circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

Question	Answer	Marks	Further Information												
3(a)	<table><thead><tr><th>current in A</th><th>number of pins</th></tr></thead><tbody><tr><td>0.25</td><td>1</td></tr><tr><td>0.50</td><td>2</td></tr><tr><td>0.75</td><td>3</td></tr><tr><td>1.00</td><td>1</td></tr><tr><td>1.25</td><td>5</td></tr></tbody></table>	current in A	number of pins	0.25	1	0.50	2	0.75	3	1.00	1	1.25	5	1	Accept any indication of the correct answer, e.g. ticking or underlining or circling the current of 1.00 A, but circling 1 pin takes precedence
current in A	number of pins														
0.25	1														
0.50	2														
0.75	3														
1.00	1														
1.25	5														
3(b)	wrapping more coils (around the core)	1	Accept more cells or greater voltage Accept use thicker wire Accept core material												

Question	Answer	Marks	Further Information
4	part description clouds of particles in space object orbiting a planet large group of stars small planets planets orbiting around a star galaxy solar system stellar dust 	3	each correct line = 1 mark if two lines from one part and one is incorrect = 0 marks for that part

Question	Answer	Marks	Further Information
5(a)	river	1	Accept a habitat that contains water, e.g. pond or lake or stream or sea or ocean
5(b)(i)	otters eat many big fish	1	Accept toxins accumulate in the body tissues (of the otter) or toxins are not broken down or toxins are not excreted
5(b)(ii)	bioaccumulation	1	Accept phonetic spellings
5(c)	loss of species or extinction of species or fewer top predators or (idea of) reduction in biodiversity	1	Accept correct example of impact on given food chain, e.g. otters die

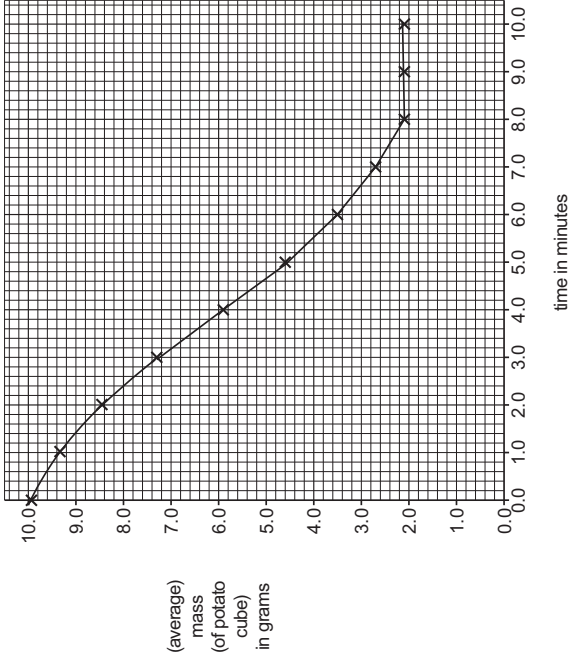
Question	Answer	Marks	Further Information
6(a)	 solution  particle diagram	1	all three lines correct for the mark
6(b)	solute solvent	1	both correct for the mark Accept soluble

Question	Answer	Marks	Further Information
7(a)	yellow	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
7(b)	(colour) blue (this is because magenta) reflects blue light or does not absorb blue light (colour) black (this is because green) absorbs blue light or does not reflect blue light	2	magenta colour and explanation both correct = 1 mark green colour and explanation both correct = 1 mark
7(c)	(colour light) green and (reason) yellow filter only lets the green through or absorbs all colours except for green or absorbs blue light	1	Note any colour other than green = 0 marks Accept cyan is green and blue, yellow is green and red (so only green passes through)

Question	Answer	Marks	Further Information										
8(a)(i)	125	1											
8(a)(ii)	the higher the (global) temperature the lower the volume (of ice on Earth)	1	Accept ora										
8(b)	<table><tr><td>average rainfall for a particular region</td><td>✓</td></tr><tr><td>changes in the atmosphere during a 24 hour period</td><td></td></tr><tr><td>highest temperature recorded in a week</td><td></td></tr><tr><td>long term temperature pattern in a given area</td><td>✓</td></tr><tr><td>predicting when the next thunderstorm will happen</td><td></td></tr></table>	average rainfall for a particular region	✓	changes in the atmosphere during a 24 hour period		highest temperature recorded in a week		long term temperature pattern in a given area	✓	predicting when the next thunderstorm will happen		2	two correct ticks = 2 marks three ticks two correct = 1 mark three ticks one correct = 0 marks more than three ticks = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
average rainfall for a particular region	✓												
changes in the atmosphere during a 24 hour period													
highest temperature recorded in a week													
long term temperature pattern in a given area	✓												
predicting when the next thunderstorm will happen													

Question	Answer	Marks	Further Information
9(a)	43.25	1	
9(b)	10	1	Accept ecf from part (a) Accept any value above 10 and below 14

Question	Answer	Marks	Further Information
10	liquids mix or intermingle or collide or spread out faster or more quickly	2	all three correct = 2 marks one or two correct = 1 mark Accept a named liquid Ignore diffuse

Question	Answer	Marks	Further Information
11(a)		4	all three points plotted correctly = 1 mark x-axis label with unit = 1 mark y-axis label and unit = 1 mark or both units correct = 1 mark or both labels correct = 1 mark curve of best fit = 1 mark Accept ecf from plots Do not accept dot to dot or straight line
11(b)	no ticked (no marks) and the potato only contained 79% water. or if the prediction was correct the potato mass would have gone down to 1 g.	1	if both boxes are ticked = 0 marks if neither box ticked accept no within the explanation
Question	Answer	Marks	Further Information
12(a)	core	1	
12(b)	magnetic field or field lines	1	

Question	Answer	Marks	Further Information
13(a)(i)	1	1	
13(a)(ii)	(idea of) measure the same volume of water each time	1	Accept (idea of) stir at same speed Ignore same mass of solid Do not accept repeat investigation or get another set of results
13(a)(iii)	unreactive or does not react	1	
13(b)	calcium oxide + water → calcium hydroxide	1	all correct for the mark Accept calcium oxide and water in either order

Question	Answer	Marks	Further Information
14(a)	(moment =) force $\times$ distance or (moment =) 500×1.5 (moment =) 750 (unit) Nm	3	Accept 750 Nm with or without working = 3 marks Accept 750 with or without working = 2 marks correct equation or substitution = 1 mark Accept (moment =) $F \times d$ or Fd correct units = 1 mark Accept Newtonmeter
14(b)	weight $\times$ distance = 750 or $600 \times d = 750$ or $d = 750 \div 600$ 1.25 (m)	2	Accept correct answer with or without working = 2 marks Accept equation or evidence of equation, e.g. correct substitution = 1 mark Accept ecf from (a) Accept 1.2 (m) or 1.3 (m)

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